CLASS 6 — MATHS · FORMULA SHEET

Chapter 8: **Playing with Constructions** | Quick revision, one page

Circle terms

- Centre O; radius r = distance from the centre to any point on the circle.
- Chord: a segment joining two points of the circle. Diameter: a chord through the centre.
- $d = 2r$, so $r = \dfrac{d}{2}$. The diameter is the longest chord.
- Point at distance x from the centre: inside if $x < r$, on the circle if $x = r$, outside if $x > r$.
- Concentric circles: same centre, different radii. Width of the ring $= R - r$.

Rectangle and square

- Rectangle: opposite sides equal, all angles 90°, diagonals equal and bisect each other.
- Square: all sides equal, all angles 90°, diagonals equal and bisect each other (and cut at right angles).
- If the diagonals meet at O: $OA = OB = OC = OD = \dfrac{1}{2} \times$ diagonal.
- Perimeter: rectangle $2(l + b)$; square $4a$. Area: rectangle $l \times b$; square a^2.

Constructions with compass and ruler

- Circle: fix the compass to the radius, put the point at the centre, turn once.
- Perpendicular bisector of AB: open the compass more than half of AB; from A and B draw arcs on both sides; join the crossing points.
- Every point on the perpendicular bisector is equidistant from A and B; it meets AB at its midpoint at 90°.

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